GOOD MODULI SPACES, POSITIVITY AND RATIONALITY

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Abstract. This course will be an introduction to the theory of good moduli spaces (in the sense of Alper). We will generalize filtrations from a vector space to an algebraic variety, define theta-reductivity, S-completeness, and unpuncturedness of inertia. Examples will include good moduli spaces for Deligne-Mumford stable curves, and K-semistable Fanos, with the focus on the cases where there is a positive tautological line bundle and the good moduli space is particularly simple, i.e., rational.

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1. Moduli of rational curves

We will begin by interpolating between $\overline{M}_{0,n}$ (which is a scheme) and the stack $(\mathbb{P}^1)^n /\!/ \mathrm{PGL}(2)$ (which has a good moduli space by definition) via Hassett spaces $\overline{M}_{0,A}$. There is a single smooth rational curve, which is $\mathbb{P}^1 = \{x = 0\} \subseteq \mathbb{P}^2_{x,y,z}$.

Viewing the curve inside $\mathbb{P}^2$, we may (see Figure 1) consider the family

$$\begin{tikzcd} \left\{xy - t^2z^2 = 0\right\} \arrow[r, equal] & \mathfrak{X} \arrow[r, hook] \arrow[d] & \mathbb{P}^2 \\ & \mathbb{A}^1_t & \end{tikzcd}$$

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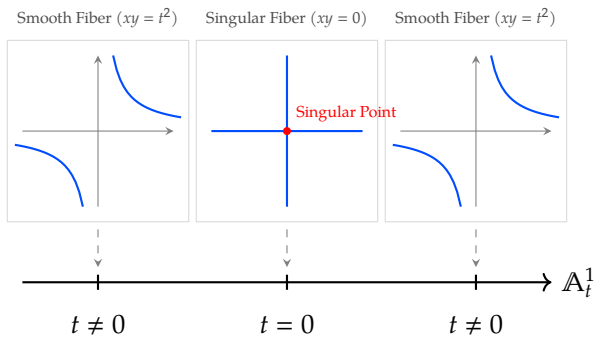


Family: $xy - t^2z^2 = 0$
In Affine Patch ($z = 1$):
$xy = t^2$

The map $\pi : \mathscr{X} \to \mathbb{A}^1_t$ is a fibration with nodal degeneration at $t = 0$.

FIGURE 1. The family $\left\{xy - t^2z^2 = 0\right\}$ over $\mathbb{A}^1$

This morphism is a (flat) isotrivial degeneration, and will be the first example of a test configuration. What this means is that there is an action of $\mathbb{G}_m$ on $\mathfrak{X}$ by the formula

$$\lambda([x, y, z], t) = ([\lambda x, \lambda y, z], \lambda t)$$

such that $\mathfrak{X} \to \mathbb{A}^1$ is equivariant. More generally, any tree of rational curves joined by nodes is also a rational curve (as in it has genus 0).

Remark 1.1. A **reduced** rational curve is a "tree" of $\mathbb{P}^1$'s meeting in m-fold rational points, where each singularity is locally analytically isomorphic to the union of the coordinate axes in $\mathbb{A}^m$.

In fact, the definition of a rational curve tells about what kinds of singularities it has, and then there is a classification of curve singularities. Another viewpoint is that if we degenerate a rational normal curve in $\mathbb{P}^m$ to a cone over a hyperplace section, we get a union of m lines meeting at a point in exactly the way described above.

We will now consider the functor $\widetilde{\mathscr{U}}_{0,n}$ of n-pointed smooth rational curves. In particular, a family $\mathfrak{X} \to B$ must have n sections $\sigma_1, \dots, \sigma_n \colon B \to \mathfrak{X}$, which are not required to be disjoint. We may now consider the larger functor $\widetilde{\mathscr{M}}_{0,n}$ of n-pointed reduced rational curves with sections σ_i avoiding the singularities of the fibers. Unfortunately, this is extremely non-separated, so we will consider stability conditions.

Define the functor $\overline{M}_{0,n}$ of n-pointed rational curves with at worst nodal singularities such that all sections $\sigma_i(B)$ are disjoint and tha4t the relative dualizing sheaf

$$\omega_{\mathfrak{X}/B}(\sigma_1 + \cdots + \sigma_n)$$

is relatively ample over B. One reason we consider this (and that it makes sense) is that the m-fold rational point is Gorenstein if and only if $m \leq 2$. This stability condition was introduced by Deligne and Mumford. In fact, the condition on the relative dualizing sheaf is equivalent to enforcing that the automorphism group of each fiber is finite, but is easier to modify for obvious reasons. From either perspective, it is clear that we must have a tree of $\mathbb{P}^1$'s with at least three special points on each component. One example can be seen in Figure 2.

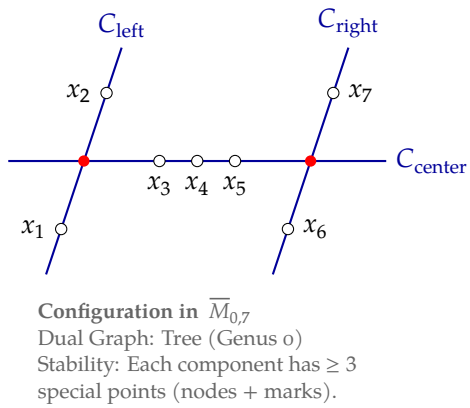


FIGURE 2. An example of a stable curve of genus 0 with marked points

One way to turn a family with an m-fold rational point into a nodal one is to blow up the singular point. The remaining conditions are non-disjoint sections, which can be separated by finitely many blowups. Finally, the unstable components can be contracted by the basic theory of surfaces (they have self-intersection either -1 or -2). In particular, we have sketched a proof that $\overline{M}_{0,n}$ is universally closed. In fact, it is also separated, but we will omit the proof here.

Suppose we have a family $\pi\colon \mathfrak{X} \to B$ in $\widetilde{\mathscr{M}}_{0,n}$. Then we have $n+1$ line bundles on B where n of them are given by

$$\sigma_i^* \omega_{\mathfrak{X}/B} = \sigma_i^*(N^\vee_{\sigma_i(B)/X})$$

(this is fine even in the non-Gorenstein case because the sections avoid the singularities) and the last is given by

$$\delta = \pi_*(c_2(\Omega^1_{\mathfrak{X}/B})),$$

which counts singularities. For example, if B is a smooth projective curve, then

$$\deg \delta = \sum_m m \cdot (\#\{\text{m-fold singularities}\}).$$

Theorem 1.2. *The line bundle*

$$\psi - \delta = \left[\sum_{i=1}^n \psi_i\right] - \delta$$

is ample on $\overline{M}_{0,n}$.

Remark 1.3. Projectivity of $\overline{M}_{0,n}$ can be proved in more synthetic ways, for example via the explicit blowup construction of Keel.

1.1. **Hassett moduli spaces.** There is a modification of this, which is the functor $\overline{M}_{0,A}$, where

$$A = (\alpha_1, \dots, \alpha_n) \in (0,1] \cap \mathbb{Q}.$$

Here, we consider n-pointed rational curves which are at worst nodal, but where the sections $\sigma_{i_1}, \dots, \sigma_{i_k}$ can meet if and only if

$$\alpha_{i_1} + \cdots + \alpha_{i_k} \leq 1$$

and the condition on the dualizing sheaf is that $\omega_{\mathfrak{X}/B}\left(\sum_i \alpha_i \sigma_i\right)$ is ample. Projectivity of this is proven using similar methods to $\overline{M}_{0,n}$.

More specifically, the functor is given by

$$\overline{M}_{0,A}(T) = \left\{ \begin{array}{c} \mathscr{C} \\ \pi \big\downarrow \big\uparrow \sigma_1, \dots, \sigma_n \\ T \end{array} \middle| \begin{array}{c} \text{fibers at worst nodal} \\ \omega_{\mathscr{C}/T}(\sum \alpha_i \sigma_i) \text{ is } \pi\text{-ample} \\ \text{sections with sum of weights} > 1 \text{ cannot meet} \end{array} \right\}.$$

This functor is actually representable by a scheme because the three conditions imply that there are no automorphisms.

Example 1.4. If $A = (1, \dots, 1)$ or more generally if the sum of any two weights is strictly greater than 1, then $\overline{M}_{0,A} = \overline{M}_{0,n}$.

Example 1.5. If $A = (1, \varepsilon, \dots, \varepsilon)$ such that $1 - (n-1)\varepsilon > 2$ and $\varepsilon \leq \frac{1}{n-2}$, then the combinatorics imply that all marked points lie on the same component. If we call the heavy marked point ∞, then all of the light markings lie in $\mathbb{A}^1$. By translation invariance, we may assume that

$$x_1 + \cdots + x_{n-1} = 0,$$

and therefore we obtain

$$[x_1 : \cdots : x_{n-1}] \in \mathbb{C}^{n-2} \smallsetminus \{0\}/\mathbb{C}^\times \simeq \mathbb{P}^{n-3}.$$

Remark 1.6. There exists a morphism $\overline{M}_{0,n} \to \overline{M}_{0,A}$ given (when the base is a curve) by changing the weights and contracting all of the (–1)-curves in the fibers where the total weight of the sections has weight at most 1.

Example 1.7. If we set $A = (\varepsilon, \dots, \varepsilon)$ such that $\varepsilon \approx \frac{2}{n} + \varepsilon'$, then the curve must be a $\mathbb{P}^1$ with n points such that we cannot have $\lceil \frac{2}{n} \rceil$ colliding. More concretely, if $n = 2m+1$ and $\varepsilon = \frac{1}{m}$, then we have a $\mathbb{P}^1$ with $2m+1$ marked points such that at most m can collide. In particular, we have

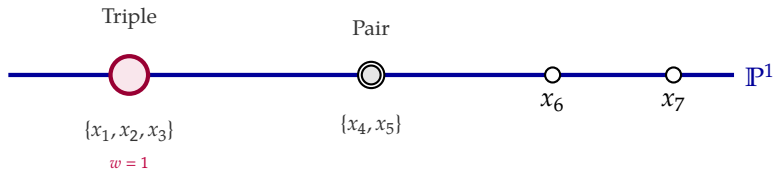
$$\begin{aligned} \overline{M}_{0,A} &= \frac{(\mathbb{P}^1)^{2m+1} \smallsetminus \left\{(m+1)\text{-fold diagonals}\right\}}{\mathrm{PGL}(2)} \\ &= ((\mathbb{P}^1)^{2m+1})^{\mathrm{ss}} \,/\!/\, \mathrm{PGL}(2), \end{aligned}$$

where the linearization is the tensor product of the $\mathscr{O}(1)$ on each factor. For an example when $m = 7$, see Figure 3.

1.2. **Semistable objects.** Now we will define a new moduli space

$$\overline{M}^*_{0,A}(T) = \left\{ \begin{array}{c} \mathscr{C} \\ \pi \big(\big\downarrow \big\uparrow \big) \sigma_1, \dots, \sigma_n \\ T \end{array} \middle| \begin{array}{c} \text{fibers at worst nodal} \\ \omega_{\mathscr{C}/T}(\sum \alpha_i \sigma_i) \text{ is } \pi\text{-nef} \\ \text{sections with sum of weights} > 1 \text{ cannot meet} \end{array} \right\}$$

by relaxing the ampleness condition. This can happen if we have sections whose weights sum to exactly 1.



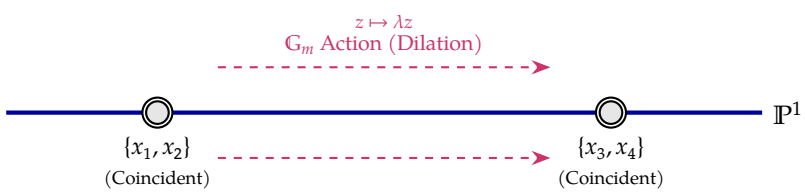
Space: $\overline{M}_{0,(\varepsilon^7)}$, $\varepsilon = \frac{1}{3}$
Topology: All stable curves are **irreducible** (smooth).
Reason: Splitting the curve requires $W_{\text{left}} > 1$ and $W_{\text{right}} > 1$.
Total weight is $\frac{7}{3}$. If $W_{\text{left}} \geq \frac{4}{3}$ (min 4 pts),
then $W_{\text{right}} \leq 1$, which is unstable.
Coincidences: Up to 3 points may coincide ($3 \times \frac{1}{3} \leq 1$).

FIGURE 3. An object in $\overline{M}_{0,\left(\frac{3}{10},\frac{3}{10},\frac{3}{10},\frac{3}{10},\frac{3}{10},\frac{3}{10},\frac{3}{10}\right)}$

Example 1.8. If $n = 2m$ and $A = \left(\frac{1}{m}, \dots, \frac{1}{m}\right)$, then we have

$$\overline{M}^*_{0,A} = (\mathbb{P}^1)^{2m} \,/\!/\, \mathrm{PGL}(2).$$

This is not a scheme and does not even have a coarse moduli space because the curve where the first m markings collide and the last m markings collide has an entire $\mathbb{G}_m$ of automorphisms. For an example when $n = 4$, see Figure 4.



Configuration: 4 points, weights $w_i = 1/2$.
Geometry: Smooth curve ($\mathbb{P}^1$) with 2 effective marked locations.
Automorphisms: Fixing 2 points on $\mathbb{P}^1$ leaves 1 degree of freedom (scaling), hence Aut $\cong \mathbb{G}_m$.
Note: This is strictly semistable if $\sum w_i = 2$.

FIGURE 4. A strictly semistable object in $\overline{M}^*_{0,\frac{1}{2}}$.

The Hassett moduli spaces have a wall-and chamber-structure, where for example, if $A = (\alpha, \dots, \alpha)$ is S_n-symmetric, then $\overline{M}_{0,A}$ is constant for $\alpha \in \left(\frac{1}{k+1}, \frac{1}{k}\right)$, but on the walls we only have $\overline{M}^*_{0,A}$. The main observation is that if we write $\overline{M}_{0,\alpha}$ for $\overline{M}_{0,(\alpha,\dots,\alpha)}$, then there are open immersions

$$\overline{M}_{0,\frac{1}{k}-\varepsilon} \subset \overline{M}^*_{0,\frac{1}{k}} \supset \overline{M}_{0,\frac{1}{k}+\varepsilon}.$$

If we consider the complement $\overline{M}^*_{0,\frac{1}{k}} \smallsetminus \overline{M}_{0,\frac{1}{k}-\varepsilon}$, then it consists of curves which contain an irreducible component which is a tree in the dual graph and contains k marked points, so this part of the moduli space has a factor which describes the rest of the curve and one which is $[\mathbb{A}^{k-2}/\mathbb{G}_m]$. More precisely, we have

$$\overline{M}^*_{0,\frac{1}{k}} \smallsetminus \overline{M}_{0,\frac{1}{k}-\varepsilon} = \overline{M}^*_{0,(1,\frac{1}{k},\dots,\frac{1}{k})} \times \overline{M}^*_{0,(1,\frac{1}{k},\dots,\frac{1}{k})}$$

This actually allows us to construct a test configuration, where if we have $\mathbb{P}^1 \times \mathbb{A}^1$ with a bunch of sections, we can blow up a point on the central fiber and then contract the central (-1)-curve with all of the sections.

Exercise 1.9. Figure out what $\overline{M}^*_{0,\frac{1}{k}} \smallsetminus \overline{M}_{0,\frac{1}{k}+\varepsilon}$ is.

For now, we will denote this picture by $\mathscr{M}^- \subset \mathscr{M} \supset \mathscr{M}^+$. Then in fact we actually have

$$\mathscr{M} \smallsetminus \mathscr{M}^+ = \overline{M}_{0,(1,\frac{1}{k},\dots,\frac{1}{k})} \times [\mathbb{C}/\mathbb{C}^\times],$$

where the second factor comes from the place where k of the sections collide. In this picture, we can degenerate out a new rational tail which has only the k colliding sections (which has a $\mathbb{C}^\times$-action), which is another example of a test configuration, as in Figure 5.

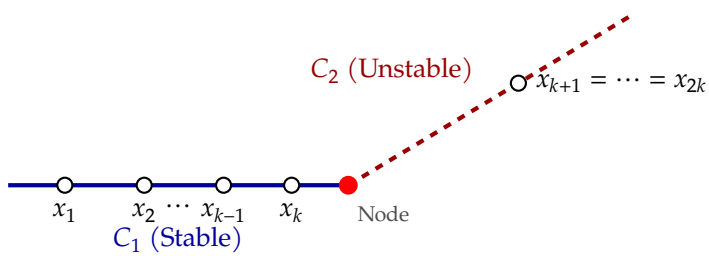


FIGURE 5. A curve that we bubble out

Later, when we discuss good moduli spaces, there is a diagram

$$\begin{tikzcd}
\mathscr{M}^+ \arrow[r, hook] \arrow[d] & \mathscr{M} \arrow[d] & \mathscr{M}^- \arrow[l, hook'] \arrow[d] \\
M^+ \arrow[r] & M & M^- \arrow[l, "\sim"']
\end{tikzcd},$$

where the rightward arrow in the bottom row is a divisorial contraction.

In fact, there is a series of divisorial contractions

$$\overline{M}_{0,n} \to \cdots \to \overline{M}_{0,A} \to \cdots \to (\mathbb{P}^1)^n \,/\!/\, \mathrm{PGL}(2)$$

and another series of divisorial contractions

$$\overline{M}_{0,n} \to \cdots \to \overline{M}_{0,A} \to \cdots \to \mathbb{P}^{n-3}.$$

1.3. **Divisors on $\overline{M}_{0,n}$.** There are boundary divisors $\Delta_{I,J}$ where $I \sqcup J = \{1, \dots, n\}$ where the curve degenerates into two components, one with the markings from I and the other with the markings from J. It is clear that

$$\Delta_{I,J} \simeq \overline{M}_{0,|I|+1} \times \overline{M}_{0,|J|+1}.$$

We will denote their union by Δ and call it the total boundary.

It is easy to see that $\overline{M}_{0,n} \smallsetminus \Delta =: M_{0,n}$ consists only of smooth curves and is given by

$$M_{0,n} = (\mathbb{C}^\times \smallsetminus 1)^{n-3} \smallsetminus \text{all diagonals}.$$

This has no Picard group, so the entire Picard group of $\overline{M}_{0,n}$ is generated by the boundary divisors. For example, $\overline{M}_{0,3}$ is a point, $\overline{M}_{0,4} = \mathbb{P}^1$, and $\overline{M}_{0,5}$ is $\mathbb{P}^1 \times \mathbb{P}^1$ blown up at three points.

Exercise 1.10. Compute the Picard rank of $\overline{M}_{0,n}$.

If we consider a 1-dimensional base B, then basic the theory of surfaces tells us that $\psi_{i|_B} = -\sigma_i^2$, and if we blow down all rational tails in $\mathscr{C}$ to form a new $\mathbb{P}^1$-bundle $\mathscr{D} \to B$ which has a section σ_1 with self-intersection $-r$ and a number of other sections $\tilde{\sigma}_i$ with self-intersection $+r$. If we now blow up the other collision points, then we see that

$$(\psi_1 + \psi_i)_{|B} = \sum_{\substack{1 \in I \\ i \in J}} \Delta_{I \sqcup J} \cdot B.$$

In particular, because $\overline{M}_{0,n}$ is smooth, we have derived Keel's relation, which is that for all $i \neq j$, then

$$\psi_i + \psi_j \sim \sum_{\substack{i \in I \\ j \in J}} \Delta_{I \sqcup J}.$$

These in fact generate all relations.

We now consider the morphism $f_{n+1} \colon \overline{M}_{0,n+1} \to \overline{M}_{0,n}$ given by deleting the last marked point and stabilizing. This is in fact the universal family over $\overline{M}_{0,n}$, so in particular it has n disjoint sections $\sigma_1, \dots, \sigma_n$. From this perspective, we can construct $\overline{M}_{0,5}$ by blowing up the points on $\mathbb{P}^1 \times \mathbb{P}^1$ where the diagonal intersects the three horizontal sections corresponding to 0, 1, and ∞. If we blow down the four (-1)-curves corresponding to the diagonal section and the three fibers that we blew up, the family becomes the pencil of conics passing through four points on $\mathbb{P}^2$ and the reducible fibers are in fact the three degenerate members (see Figure 6). In particular, we see that $\mathrm{Pic}(\overline{M}_{0,5}) = \mathbb{Z}^5$.

Another way to see this is to consider the sequence

$$\overline{M}_{0,n} \simeq \overline{M}_{0,(\frac{1}{2},\dots,\frac{1}{2})} \to \overline{M}_{0,(\frac{1}{3},\dots,\frac{1}{3})} \to \cdots \to (\mathbb{P}^1)^n \mathbin{/\!/} \mathrm{PGL}(2)$$

and use this to compute the Picard rank, which is

$$\rho(\overline{M}_{0,n}) = 2^{n-1} - \binom{n}{2} - 1.$$

A more precise description is that

$$\mathrm{Pic}(\overline{M}_{0,n}) = \mathbb{Z}\Big\langle \Delta_{I \sqcup J}, \psi_1, \dots, \psi_n \Big\rangle / \mathrm{Keel}.$$

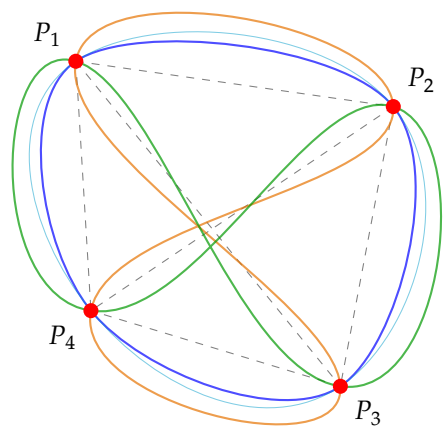


FIGURE 6. Pencil of conics through four points in general position on $\mathbb{P}^2$. The three degenerate conics correspond to pairs of lines through these points (dashed).

The blowup construction (really the one that starts with $\mathbb{P}^{n-3}$) proves that the Picard group also has no torsion.

One way to find a basis for the Picard group is the following. We will call $\Delta_{I \sqcup J}$ *non-adjacent* if I is not an arc when we arrange $1, \dots, n$ in a circle.

Proposition 1.11. *Non-adjacent boundary divisors form a basis of* $\mathrm{Pic}(\overline{M}_{g,n})$.

Proof. We use Keel's relations to rewrite every divisor as a linear combination of non-adjacent ones, and then we check that the number of non-adjacent ones is exactly the Picard rank.

For example, if we have $I = \{1, 2\}$, we will use the relations for $\psi_1 + \psi_2$, $\psi_3 + \psi_n$ (with a plus sign), $\psi_1 + \psi_n$, and $\psi_2 + \psi_3$ (with a minus sign), then the left hand side vanishes while the only adjacent boundary divisor remaining is the one with $I = \{1, 2\}$. The general case of this is left as an exercise, as is counting the non-adjacent boundary divisors. □

We have discussed the fact that $\overline{M}_{0,n}$ is a projective variety. We know it is proper, but we would like to produce an ample line bundle. One candidate is

$$\psi - \Delta := \psi_1 + \cdots + \psi_n - \sum \Delta_{I \sqcup J}.$$

The first step to proving that this is ample is to sum all of Keel's relations, which gives us

$$(n-1)\psi = |I||J|\Delta_{I \sqcup J}.$$

Because $|I||J| \geq 2(n-2) > n-1$ (here $n \geq 4$), we see that

$$\Psi - \Delta = \sum c_{I \sqcup J} \Delta_{I \sqcup J},$$

where all coefficients are positive.

We will now use the ampleness criterion and show that $(\psi - \Delta).B > 0$ for any irreducible proper curve $B \hookrightarrow \overline{M}_{0,n}$. If B intersects the interior nontrivially, then by definition it has to intersect the boundary with positive degree. The other case is when $B \hookrightarrow \Delta_{I \sqcup J}$ lives inside one of the boundary divisors.

In this case, we have

$$B \hookrightarrow \Delta_{I \sqcup J} \simeq \overline{M}_{0,k+1} \times \overline{M}_{0,n-k+1}.$$

Restricting $\psi - \Delta$, we obtain

$$(\psi - \Delta)_{|\Delta_{I \sqcup J}} \simeq (\psi - \Delta)_{\overline{M}_{0,k+1}} \boxtimes (\psi - \Delta)_{\overline{M}_{0,n-k+1}},$$

so we induct using the fact that on $\overline{M}_{0,4} \simeq \mathbb{P}^1$, $\psi - \Delta = \mathscr{O}(1)$.

2. Geometric invariant theory

2.1. **Groupoids and coarse moduli.** We will now recall the Yoneda embedding, which is the fully faithful functor

$$\mathsf{Sch}_S \to \mathsf{Fun}(\mathsf{Sch}_S^{\mathrm{op}}, \mathsf{Set}), \qquad X \mapsto h_X = \mathrm{Mor}_{\mathsf{Sch}_S}(-, X).$$

Definition 2.1. A scheme G is a *group scheme* if the functor h_G factors through the embedding $\mathsf{Grp} \hookrightarrow \mathsf{Set}$.

Example 2.2.

(1) There is the group scheme $\mathbb{G}_m = S \times \mathbb{Z}[x, x^{-1}]$ which sends a scheme T to $H^0(T, \mathscr{O}_T)^\times$.
(2) There is the group scheme $\mathbb{G}_a = S \times \mathbb{Z}[x]$ which sends a scheme T to $H^0(T, \mathscr{O}_T)$ with addition.

Definition 2.3. A scheme G is a *group scheme* if there exist morphisms

$$\mu \colon G \times_S G \to G, \qquad \beta \colon G \to G, \qquad e \colon S \to G$$

satisfying the usual group axioms:

(1) The diagram

$$\begin{tikzcd} G \times G \times G \arrow[r, "1\times\mu"] \arrow[d, "\mu\times 1"] & G \times G \arrow[d, "\mu"] \\ G \times G \arrow[r, "\mu"] & G \end{tikzcd}$$

commutes (associativity);

(2) The diagram

$$\begin{tikzcd} G \arrow[r, "\Delta"] \arrow[d] & G \times G \arrow[r, shift left, "1\times\beta"] \arrow[r, shift right, "\beta\times 1"'] & G \times G \arrow[d, "\mu"] \\ S \arrow[rr, "e"] & & G \end{tikzcd}$$

commutes (inverse axiom).

(3) The diagram

$$\begin{tikzcd}
 & G \times_S S \arrow[dr, "1\times e"] & & \\
G \arrow[ur] \arrow[dr] \arrow[rrr, bend left, "1"] & & G \times_S G \arrow[r, "\mu"] & G \\
 & S \times_S G \arrow[ur, "e\times 1"] & &
\end{tikzcd}$$

commutes (identity axiom).

Example 2.4. If k is algebraically closed, then a (Zariski)-closed subgroup of $\mathrm{GL}_n(k)$ is a group scheme over k, and is known as a *linear algebraic group*.

Definition 2.5. An *algebraic group* over k is a smooth group scheme over k, and if it is affine, we call it *linear*.

Definition 2.6. An *action* of an algebraic group G on a scheme X is a morphism $\sigma\colon G \times X \to X$ such that the diagram

$$\begin{tikzcd}
G \times G \times X \arrow[r, "\mu\times 1"] \arrow[d, "1\times\sigma"] & G \times X \arrow[d, "\sigma"] \\
G \times X \arrow[r, "\sigma"] & X
\end{tikzcd}$$

commutes and the diagram

$$\begin{tikzcd}
X \arrow[r, "\simeq"] \arrow[rrr, bend right, "1_X"'] & S \times X \arrow[r, "e\times 1"] & G \times X \arrow[r, "\sigma"] & X
\end{tikzcd}$$

commutes.

On k-points over an algebraically closed field, this says that $g(g'(x)) = (gg')(x)$ and that $e \cdot x = x$ which are the usual group action axioms.

Construction 2.7 (Action groupoid)**.** Suppose that G acts on X. Define $R := G \times X$, $s := \mathrm{pr}_2\colon R \to X$ (source morphism), and $t := \sigma\colon R \to X$ (target morphism). Then set $j := (s,t)\colon R \to X \times X$. There is a morphism

$$c\colon R \times_{(s,t)} R \simeq G \times G \times X \xrightarrow{\mu\times 1} G \times X \cong R$$

giving composition. Finally there is the morphism

$$e\colon X \simeq S \times X \xrightarrow{e\times 1} G \times X \simeq R$$

and the reflexivity morphism

$$i\colon R \xrightarrow{\beta\times 1_X} R \simeq G \times X.$$

This construction gives a groupoid in schemes called the *action groupoid* associated to the action of G on X.

More generally, if we replace $G \times X$ by an arbitrary R, we obtain the general notion of a *groupoid* in the category of schemes. Our goal will be to prove the following theorems.

Theorem 2.8. *Let $j\colon R \to X \times X$ be a flat groupoid such that the stabilizer $j^{-1}(\Delta_X) \to \Delta_X \simeq X$ is finite. Then there exists an algebraic space "quotient" X/R and a quotient morphism $q\colon X \to X/R$ which is a uniformly categorical and geometric quotient. Moreover, if j is finite, then X/R is separated.*

Definition 2.9. Let $\sigma\colon G \times X \to X$ be a group action. Then $\phi\colon X \to Y$ is a *geometric quotient* if the square in the diagram

$$\begin{tikzcd}
T \times X \arrow[r, "\sigma"] \arrow[d, "\mathrm{pr}_2"] & X \arrow[d, "\phi"] \arrow[ddr, bend left, "\psi"] & \\
X \arrow[r, "\phi"] \arrow[drr, bend right, "\psi"'] & Y \arrow[dr, dashed, "\exists!"] & \\
& & Z
\end{tikzcd}$$

commutes and the dashed arrow is unique for any commutative diagram where the ψ exists. The categorical quotient is *uniform* if it is preserved by flat base change and *universal* if it is preserved by arbitrary base change.

Naively, we may form a quotient pre-sheaf

$$X/R\colon \mathrm{Sch}_S^{\mathrm{op}} \to \mathrm{Set}, \qquad T \mapsto X(T)/\sim,$$

where $x \sim x'$ if there exists $g \in G(T)$ such that $g \cdot x = x'$. However, this is not a sheaf in general (and is definitely not a scheme), so we need to improve our constructions.

2.2. **Good quotients.** Let $S = \operatorname{Spec} k$ for an algebraically closed field k and G be an algebraic group over k. Suppose that X is a scheme with an action

$$\sigma\colon G \times X \to X$$

of G. Now suppose we have another action

$$\sigma'\colon G \times X' \to X'$$

on another scheme X'.

Definition 2.10. A morphism $f\colon X \to X'$ is *G-invariant* if the diagram

$$\begin{tikzcd}
G \times X \arrow[r, "\sigma"] \arrow[d, "1\times f"'] & X \arrow[d, "f"] \\
G \times X' \arrow[r, "\sigma'"] & X'
\end{tikzcd}$$

commutes.

In the finite-type case, this reduces to the equation

$$g \circ g(x) = f(g \circ x)$$

for all k-points $g \in G(k)$ and $x \in X(k)$.

Definition 2.11. If G acts trivially on Y, then a G-equivariant $f\colon X \to Y$ is *G-invariant*.

In particular, f has to collapse all orbits of G to points.

Definition 2.12. A morphism $\phi\colon X \to Y$ is a *categorical quotient* of X by G if it is G-invariant and for any other G-invariant morphism ψ, there exists a unique $g\colon y \to y'$ such that $\psi = g \circ \phi$, or in other words the dashed arrow in

$$\begin{tikzcd} X \arrow[r, "\phi"] \arrow[dr, "\psi"'] & Y \arrow[d, dashed, "g"] \\ & Y' \end{tikzcd}$$

exists and is unique.

Exercise 2.13. If X has property $\mathscr{P}$ being one of normal, irreducible, reduced, connected, or locally factorial, and if $\phi\colon X \to Y$ is a categorical quotient, then Y also has $\mathscr{P}$.

Remark 2.14. If G acts on X and $U \subset X$ is a G-invariant open subset, then the group $G(k)$ acts on $\mathscr{O}_X(U)$, which is a k-algebra.

Definition 2.15. Suppose that G acts on X. A morphism $\pi\colon X \to Y$ is a *good quotient* if

(1) It is G-invariant;
(2) π is affine and surjective;
(3) If $U \subset Y$ is open, the morphism of rings $\mathscr{O}_Y(U) \to \mathscr{O}_X(\pi^{-1}(U))^G$ is an isomorphism;
(4) If W_1 and W_2 are disjoint G-invariant closed subsets in X, then $\pi(W_1) \cap \pi(W_2) = \varnothing$ and the two images are closed.

Proposition 2.16. *Every good quotient is a categorical quotient.*

Proof. Suppose that $\pi\colon X \to Y$ is a good quotient and $\phi\colon X \to Z$ is G-invariant. We will construct a morphism $g\colon Y \to Z$ filling in the dashed arrow in

$$\begin{tikzcd} X \arrow[r, "\pi"] \arrow[dr, "\phi"'] & Y \arrow[d, dashed, "g"] \\ & Y'. \end{tikzcd}$$

Let $\{V_i\}$ be a cover of Z by affine open subschemes.

The first claim we will prove is that there exists an open $U_i \subset Y$ such that $\pi^{-1}(U_i) = \phi^{-1}(V_i)$. If $x \in V_i$ is a closed point and we set $W = Z \smallsetminus V_i$, then $\phi^{-1}(x)$ and $\phi^{-1}(W)$ are disjoint G-invariant closed subsets of X. This is true for all $x \in V_i$, so we see that $\phi^{-1}(V_i) = \pi^{-1}(U_i)$ for some $U_i \subset X$. Because $Y \smallsetminus U_i = \pi(\phi^{-1}(W))$ is closed, U_i must be open.

We now treat U_i as a subset of Y. We need to construct a morphism $g_i\colon U_i \to V_i = \operatorname{Spec} A_i$. There is a map

$$A_i \to \mathscr{O}_X(\phi^{-1}(V_i))^G = \mathscr{O}_X(\pi^{-1}(U_i))^G \simeq \mathscr{O}_Y(U_i),$$

which is the desired morphism. These morphisms glue, so we are done. □

We will denote good quotients by $X /\!\!/ G$.

Lemma 2.17. *If $\pi\colon X \to X /\!\!/ G$ is a good quotient, then for all closed $y \in X /\!\!/ G$, the fiber $\pi^{-1}(y)$ contains a unique closed G-orbit.*

Proof. Consider the orbit $G \cdot x \subset \pi^{-1}(y)$ of the smallest dimension. Then $\overline{G \cdot x} \smallsetminus G \cdot x$ is a union of orbits, so it must be empty. □

Corollary 2.18. *The set $X /\!\!/ G(k)$ is in bijection with the set of closed orbits in X.*

Definition 2.19. A good quotient is *geometric* if all orbits of closed points are closed. Equivalently, $X /\!\!/ G(k)$ is in bijection with the set of $G(k)$-orbits in $X(k)$. A third equivalent definition is that $\overline{G \cdot x_1} \cap \overline{G \cdot x_2} = \varnothing$ for all $x_1, x_2 \in X(k)$ in disjoint orbits.

Lemma 2.20. *Suppose that $f\colon X \to Y$ is finite and G-equivariant. Then $G \cdot x$ is closed if and only if $G \cdot f(x)$ is closed.*

Proof. Note that equivariance implies that $G \cdot f(x) = f(G \cdot x)$. If $G \cdot x$ is closed, then so is $G \cdot f(x)$ by finiteness. If $G \cdot x$ is not closed and there exists $x' \in \overline{G \cdot x} \smallsetminus G \cdot x$ but $G \cdot f(x)$ is closed, then $\dim G \cdot x' < \dim G \cdot x$. The sequence of containments

$$G \cdot f(x') = f(G \cdot x') \subset f(\overline{G \cdot x}) = \overline{G \cdot f(x)} = G \cdot f(x)$$

implies that $G \cdot f(x') = G \cdot f(x)$, contradicting the dimension inequality. □

Definition 2.21. Let X be a G-scheme. A G-invariant subset $U \subseteq X$ is *saturated* if for all $x \in U$ and $y \in X$ such that $\overline{G \cdot x} \cap \overline{G \cdot y} \neq \varnothing$, then $y \in U$.

Example 2.22. Suppose we have $f_1, \dots, f_n \in \mathscr{O}_X(X)^G$. Then the open set

$$D(f_1, \dots, f_n) = D(f_1) \cap \cdots \cap D(f_n)$$

is saturated.

Exercise 2.23. Suppose that $\pi\colon X \to X /\!\!/ G$ is a good quotient. Then a G-invariant subset $U \subseteq X$ is saturated if and only if $U = \pi^{-1}(\pi(U))$.

One major statement we want to prove is the following.

Theorem 2.24 (Affine GIT)**.** *Suppose that G is a linearly reductive group acting on an affine scheme $X = \operatorname{Spec} R$. Then*

$$\pi\colon X \to Y = \operatorname{Spec} R^G$$

is a universal good quotient.

Definition 2.25. A group scheme G is *linearly reductive* if every finite-dimensional rational representation of G is completely reducible.

Remark 2.26. In characteristic zero, a group G is reductive (equivalently, linearly reductive), if and only if $BG = [\bullet/G] \to \bullet$ is a good moduli space.

Definition 2.27. Let V be a finite-dimensional G-representation. Consider the vector space

$$V^G = \left\{ v \in V \mid g \cdot v = v \text{ for all } g \in G \right\}.$$

By complete reducibility, there is a direct sum decomposition $V = V^G \oplus V'$, so we define a map

$$\rho\colon V \to V^G$$

by projecting onto the first factor. This is clearly G-equivariant, surjective, and idempotent, and is called the *Reynolds operator*.

By Schur's lemma, the diagram

$$\begin{tikzcd} V \arrow[r, "\psi"] \arrow[d, "\rho"] & W \arrow[d, "\rho"] \\ V^G \arrow[r] & W^G \end{tikzcd}$$

commutes.

By gluing Reynolds operators on finite-dimensional G-representations together, we obtain a Reynolds operator on infinite-dimensional rational representations, and in particular on k-algebras R as

$$\rho\colon R \to R^G.$$

Remark 2.28. Note that R^G is a k-subalgebra of R.

Let G be a linear algebraic group. Then if $G = \operatorname{Spec} S$ (note here that S is a Hopf algebra), an action

$$G \times \operatorname{Spec} R \to \operatorname{Spec} R$$

of G on an affine scheme $\operatorname{Spec} R$ corresponds to a coaction

$$R \to R \otimes_k S.$$

Composing this with with $g^*\colon S \to k$ corresponding to $g \in G(k)$, we obtain an action

$$g^*\colon R \to R.$$

This defines a k-linear action

$$G(k) \times R \to R \qquad (g, f) \mapsto g^*(f).$$

Definition 2.29. $G(k)$ acts *rationally* on R if every element of R lies in a finite-dimensional $G(k)$-invariant subspace of R.

Lemma 2.30. *The action $G(k) \times R \to R$ coming form the action of G on* $\operatorname{Spec} R$ *is rational.*

Idea of proof. For any $r \in R$, consider the coaction

$$r \in R \mapsto \sum_i r_i \otimes s_i \in R \otimes_k S.$$

Then r lies in the span of the r_i, which is finite-dimensional and $G(k)$-invariant. □

This clarifies what we meant by gluing Reynolds operators together. By complete reducibility, we can write $R = R^G \oplus R'$.

Claim 2.31. *R' is an R^G-module and the direct sum decomposition holds at the level of R^G-modules.*

Proof. We need to prove that $R^G R' \subseteq R'$. Choose a finite-dimensional subrepresentation $V \subseteq R'$ and $r \in R'$. We want to prove that $rV \subseteq R'$. Now consider the sequence

$$V \xrightarrow{r\cdot} rV \xrightarrow{\rho} (rV)^G \subseteq R^G.$$

Because V is a direct sum of nontrivial representations and R^G is a trivial representation, the composition is zero by Schur's lemma. □

Now consider the Cartesian diagram

$$\begin{tikzcd} \operatorname{Spec} T \otimes_{R^G} R \arrow[r] \arrow[d] & \operatorname{Spec} R \arrow[d] \\ \operatorname{Spec} T \arrow[r] & \operatorname{Spec} R^G. \end{tikzcd}$$

Corollary 2.32. *For all R^G-algebras T, we have*

(1) *A canonical isomorphism $(T \otimes_{R^G} R)^G \cong T$.*
(2) *Suppose that $\{I_i\}$ are G-invariant ideals of R. Then*

$$\left(\sum I_i\right) \cap R^G = \sum (I_i \cap R^G) = \sum I_i^G.$$

Proof.

(1) Because $R \simeq R^G \oplus R'$ as R^G-modules, we have

$$T \otimes_{R^G} R \simeq T \oplus (T \otimes_{R^G} R').$$

Applying the Reynolds operator, we see that

$$(T \otimes_{R^G} R)^G = \rho(T) \oplus (T \otimes_{R^G} \rho(R')) = T.$$

(2) We use the fact that $\rho(\sum I_i) = \sum \rho(I_i)$. □

Proof of Theorem 2.24. The morphism is clearly G-invariant and affine. We will begin with surjectivity. Choose $T = k$, so we are looking at a k-point of $\operatorname{Spec} R^G$. We only need to prove that $k \otimes_{R^G} R$ is nonzero, but we already know that $(k \otimes_{R^G} R)^G = k$ is nonzero. In particular, this means that every fiber is nonempty.

Now let $\operatorname{Spec} T$ be an open affine in $\operatorname{Spec} R^G$. Now we see that

$$\mathscr{O}_Y(T) = T = (T \otimes_{R^G} R)^G = \pi_* \mathscr{O}_X(T),$$

and this holds for all open affines, so it extends to an isomorphism

$$\mathscr{O}_Y \simeq (\pi_* \mathscr{O}_X)^G$$

of sheaves.

Now let $W_i = V(I_i)$ be disjoint G-invariant closed subsets of X. Then I_1 and I_2 are G-invariant ideals, and in particular

$$\overline{\pi(W_i)} = V(I_i \cap R^G).$$

This implies that

$$\overline{\pi(W_1)} \cap \overline{\pi(W_2)} = \overline{\pi(W_1 \cap W_2)},$$

so the closures are disjoint. Finally, if we choose $W_2 = \pi^{-1}(\mathrm{pt})$, we see that $\pi(W_1)$ is closed. □

Now if we have a G-invariant ideal $I \subseteq R$, the sequence

$$0 \to I^G \to R^G \to R^G/I^G \to 0$$

is also exact. In particular, if we have a G-invariant closed subscheme $Z \hookrightarrow X$, then $Z \to \pi(Z)$ is also a good quotient and the diagram

$$\begin{tikzcd} Z \arrow[r, hook] \arrow[d] & X \arrow[d] \\ \pi(Z) \arrow[r, hook] & Y \end{tikzcd}$$

commutes.

Lemma 2.33. *If $X = \operatorname{Spec} R$ is Noetherian, then so is $Y = X /\!/ G = \operatorname{Spec} R^G$.*

Proof. If $W \subseteq Y$ is closed, then $\pi(\pi^{-1}(W)) = W$, so ascending chains of ideals downstairs correspond to ones upstairs. Here, we use the fact that

$$(R/RJ)^G = ((R^G/J) \otimes_{R^G} R)^G = R^G/J$$

and thus $(RJ)^G = J$ for any ideal $J \subseteq R^G$. □

Proposition 2.34. *If R is a finitely-generated k-algebra, then so is R^G. Equivalently, if X is finite type, then so is $X /\!/ G$.*

Proof. We first consider the case that

$$R = \bigoplus_{d \geq 0} R_d$$

is a graded ring with $R_0 = k$ and G preserves the grading. In this case, $R^G \subseteq R$ is also graded and is Noetherian, and is thus finitely generated.

In the second case, G acts rationally on R. Suppose that R is generated by $x_1, \dots, x_n$. Then the generators are contained in some finite-dimensional G-subrepresentation $V \subseteq R$, so conclude using the diagram

$$\begin{tikzcd} A = \operatorname{Sym}^* V \arrow[r, two heads] \arrow[d] & R \arrow[d] \\ A^G \arrow[r, two heads] & R^G \end{tikzcd}$$

to show that R^G is finitely-generated. □

2.3. **Numerical criteria for stability.** Now suppose that $\sigma \colon G \times X \to X$ is a group action. Consider the morphism

$$\psi = (\sigma, \mathrm{pr}_2) \colon G \times X \to X \times X \qquad \psi(g, x) = (gx, x).$$

If $j_x \colon \operatorname{Spec} k \to X$ is the inclusion of a point, then consider the Cartesian diagram

$$\begin{tikzcd} G \times X \arrow[r, "\psi"] & X \times X \\ G \times \operatorname{Spec} k \arrow[r, "\psi_x"] \arrow[u, "1_G \times j_x"] & X \times \operatorname{Spec} k. \arrow[u] \end{tikzcd}$$

This defines a morphism $\psi_x \colon G \to X$ and $G \cdot x = \mathfrak{I}(\psi_x)$ is the orbit of x.

Definition 2.35. The action σ is called:

(1) *closed* if all maps ψ_x are closed;
(2) *separated* if ψ itself is closed;

(3) *proper* if ψ is proper;
(4) *free* if ψ is a closed embedding.

If we consider the Cartesian diagram

$$\begin{tikzcd} G \times X \arrow[r, "\psi"] & X \times X \\ I_X \arrow[u] \arrow[r] & X, \arrow[u, "\Delta"'] \end{tikzcd}$$

then in fact the fibers of $I_X \to X$ are the stabilizers of points in X.

Lemma 2.36. *If σ is proper, then all ψ_x are proper. This further implies that all orbits are closed and all stabilizers are proper.*

Remark 2.37. There are examples where the converse is false, even when all orbits are closed and stabilizers are trivial.

Proof. Use the valuative criterion of properness. □

Definition 2.38 (1-PS)**.** A 1-*parameter subgroup* of an algebraic group G is a morphism

$$\lambda \colon \mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}] \to G.$$

Example 2.39. If $G = \mathrm{GL}_n$, then one example of a 1-parameter subgroup is the set of matrices of the form

$$\begin{pmatrix} t^{a_1} & & \\ & \dots & \\ & & t^{a_n} \end{pmatrix}$$

in some basis.

Proposition 2.40. *An action of a* ***reductive*** *group on a finite type and separated*[1] *X over k is proper if and only if for every nontrivial* 1*-parameter subgroup*

$$\lambda \colon \mathbb{G}_m \to G,$$

the induced action of $\mathbb{G}_m$ on X is proper.

Theorem 2.41 (Iwahori)**.** *For a* 1*-PS $\lambda \colon \mathbb{G}_m \to G$ and a DVR R with fraction field K, denote by $\langle \lambda \rangle_K$ the K-point*

$$\operatorname{Spec} K \to \operatorname{Spec} k[t, t^{-1}] \to G$$

induced by sending t to a uniformizer π of R. Then

$$G(K) = \bigcup_{\lambda} G(R) \langle \lambda \rangle_K G(R)$$

is a union of double cosets.

Theorem 2.42. *If $X = \operatorname{Spec} R$ is finite type and G is linearly reductive, then the action is proper if and only if all stabilizers are finite.*

Remark 2.43. Note that finite stabilizers imply that all orbits are closed.

Corollary 2.44. *If $X /\!/ G = X/G$ is a geometric good quotient by a reductive group, then the action is proper.*

[1] Note that in Mumford's book, all schemes are assumed to be schemes and non-separated schemes are referred to as preschemes.

Remark 2.45. If $\pi\colon X \to Y$ is G-invariant and surjective then the following are equivalent:

(1) $\mathfrak{I}(\psi) = X \times_Y X \subset X \times X$;
(2) All fibers of π are orbits.

In particular, if this holds and Y is separated, then the diagram

$$\begin{tikzcd} X \times_Y X \arrow[r] \arrow[d] & X \times X \arrow[d] \\ Y \arrow[r, hook, "\Delta"] & Y \times Y \end{tikzcd}$$

is Cartesian, so ψ is separated.

Proof of Proposition 2.40. We apply the valuative criterion. Let R be a DVR with fraction field K. Consider the diagram

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r, "(\phi,\zeta)"] \arrow[d] & G \times X \arrow[d] \\ \operatorname{Spec} R \arrow[r] & X \times X \end{tikzcd}$$

Then we know that

$$\psi((\phi,\zeta)) = (\phi\zeta, \zeta) \in X(R) \times X(R)$$

by assumption. Also consider the Iwahori decomposition

$$\phi = \psi_1 \langle\lambda\rangle \psi_2$$

for $\psi_i \in G(R)$. Now consider a 1-PS $\lambda\colon \mathbb{G}_m \to G$. Then we compute

$$\begin{aligned} \psi(\langle\lambda\rangle, \psi_2\zeta) &= (\langle\lambda\rangle\psi_2\zeta, \psi_2\zeta) \\ &= (\psi_1^{-1}\phi\zeta, \psi_2\zeta) \in X(R) \times X(R). \end{aligned}$$

Therefore the morphism $\mathbb{G}_m \times X \to X \times X$ is proper, so $\langle\lambda\rangle \in \mathbb{G}_m(R) \subset G(R)$. □

Proposition 2.46. *Suppose that $\mathbb{G}_m$ acts on X such that the action is induced from an action of $\mathbb{G}_m$ on $\mathbb{P}^n_k$ and a locally closed $\mathbb{G}_m$-equivariant embedding $X \hookrightarrow \mathbb{P}^n_k$. Then the action is proper if and only if all stabilizers are finite.*

Proof of Theorem 2.42. If a good quotient exists and all orbits are closed with finite stabilizers, then $X /\!/ G = X/G = \operatorname{Spec} R^G$ is geometric. Therefore, σ is separated, which implies that all 1-PS actions are proper, and thus that σ is proper. □

Proof of Proposition 2.46. $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$ acts on $\mathbb{P}^n = \operatorname{Proj} k[x_0, \dots, x_n]$ by the $x_i \mapsto t^{\alpha_i} x_i$ for $\alpha_i \in \mathbb{Z}$. Choose $R = k[\![t]\!]$ and $K = k(\!(t)\!)$. Now the calculation with the diagram

$$\begin{tikzcd} \mathbb{G}_m \times X \arrow[d] & \operatorname{Spec} K \arrow[l, "(\phi,\zeta)"] \arrow[d] \\ X \times X & \operatorname{Spec} R \arrow[l] \end{tikzcd}$$

becomes very explicit. We assume that $\phi\zeta, \zeta \in X(R) \subset \mathbb{P}^n(R)$. Write $\zeta = [a_0 : \cdots : a_n] \in \mathbb{P}^n(R)$ for $a_i \in R$. Then we see that

$$\phi\zeta = [A^{\alpha_i} a_i] \in \mathbb{P}^n(R),$$

where $\phi = A \in K^\times = \mathbb{G}_m(K)$. Because $(\phi\zeta, \phi) \in \Im(\psi)$, so is $(\phi\zeta(0), \zeta(0))$ by separatedness. Therefore, $(\phi\zeta)(0) = \phi_0(\zeta(0))$ for some $\phi_0 \in \mathbb{G}_m(k) = k^\times$. Now there exists some $\tilde{\phi} \in \mathbb{G}_m(R)$ such that $\tilde{\phi}(0) = \phi_0$. Replace $\zeta \in X(R)$ by $\tilde{\zeta} = \tilde{\phi}^{-1}\zeta \in X(R)$. Therefore,

$$\phi\tilde{\zeta}(0) = \tilde{\zeta}(0)$$

and thus $(\phi\tilde{\zeta}, \tilde{\zeta}) \in X(R) \times X(R)$ satisfies

$$(\phi\tilde{\zeta}(0), \tilde{\zeta}(0)) \in \Delta_X \subset X(k) \times X(k).$$

Now write

$$\phi\tilde{\zeta} = [A^{\alpha_0}\tilde{a}_0 : \cdots : A^{\alpha_n}\tilde{a}_n] \in \mathbb{P}^n_R.$$

Clearly, we have $\tilde{a}_i(0) = A^{\alpha_i}\tilde{a}_i(0)$, so if $\tilde{a}_i \neq 0$ and $\alpha_i \neq 0$, then A is a unit. But the point

$$[\tilde{a}_0 : \cdots : \tilde{a}_n] \in \mathbb{P}^n_k$$

has finite stabilizer under the action with weights $\alpha_0, \dots, \alpha_n$, so the desired i exists. □

Example 2.47 (Non-proper action, Mumford)**.** Let $G = \mathrm{SL}(2)$ and

$$X = \left\{(F_1, F_4) \mid F_1 \neq 0, F_4 = Q^2, Q \text{ quadric}, \det(Q) = 1\right\} \subset \mathbb{C}^2 \times \mathrm{Sym}^4(\mathbb{C}^2).$$

The action will be induced from the natural action of SL(2) on $\mathbb{C}^2$.

Exercise 2.48. There is a map

$$\pi\colon X \to \mathbb{A}^1 \qquad \pi(\alpha X + \beta Y, Q^2) = Q(-\beta, \alpha) \in \mathbb{C}$$

which is SL(2)-invariant with $\pi^{-1}(\pi(x)) = \mathrm{SL}(2) \cdot x$. Moreover, all stabilizers are trivial.

There is a locus

$$Y = \left\{\begin{pmatrix} 0 & \lambda \\ -\lambda^{-1} & 0 \end{pmatrix}, (\lambda x + y, x^2y^2)\right\} \hookrightarrow \mathrm{SL}(2) \times X,$$

which is closed. Then we see that

$$\begin{aligned}\Psi(Y) &= (\sigma, \mathrm{pr}_2)(Y) \\ &= \left\{(-\lambda x + y, x^2y^2), (\lambda x + y, x^2y^2)\right\} \subset X \times X,\end{aligned}$$

which is not closed. Therefore, the orbit space map cannot be any kind of good quotient.

This example is morally the same the example of the action of $\mathbb{G}_m \rtimes \mathbb{Z}_2$ on $\mathbb{A}^2 \smallsetminus 0$, where $\mathbb{G}_m$ acts with weights 1 and –1 and $\mathbb{Z}_2$ acts by swapping the coordinates. This has a map to $\mathbb{A}^1$ which separates orbits, but this is still not a good quotient because even though the diagonal Δ of $(\mathbb{A}^2 \smallsetminus 0) \times (\mathbb{A}^2 \smallsetminus 0)$ is closed, $\Psi(\Delta)$ is not closed.

In the language of stacks, in fact thre is an isomorphism

$$[X/\mathrm{SL}(2)] \simeq [(\mathbb{A}^1 \text{ with doubled origin})/\mathbb{Z}_2],$$

where $\mathbb{Z}_2$ acts by swapping the two origins. This stack does not have a good moduli space which is a scheme, but it does have a good moduli space which is a non-separated algebraic space by Keel-Mori.

Example 2.49 (Hironaka)**.** There is a proper but non-projective threefold X with a free action of $\mathbb{G}_m$ which is not proper, and the quotient is not a scheme but is an algebraic space.

2.4. **Linearizations and GIT.** Let X be finite type and separated over an algebraically closed field k, $\mathscr{L}$ be a line bundle on X, and $\sigma\colon G \times X \to X$ be an action.

Definition 2.50. A *G-linearization* of $\mathscr{L}$ is an action

$$\Sigma\colon G \times \mathbb{L} \to \mathbb{L}$$

on the total space of $\mathscr{L}$ such that the diagram

$$\begin{tikzcd} G \times \mathbb{L} \arrow[r, "\Sigma"] \arrow[d, "1_G \times \pi"] & \mathbb{L} \arrow[d, "\pi"] \\ G \times X \arrow[r, "\sigma"] & X \end{tikzcd}$$

commutes and the induced map

$$G \times \mathbb{L} \to \sigma^* \mathbb{L} = (G \times X) \times_X \mathbb{L}$$

is a morphism of line bundles over $G \times X$.

Example 2.51. If G acts on V rationally by some morphism $G \to \mathrm{GL}(V)$, then the induced action on $\mathbb{P}(V)$ comes with a linearization of $\mathscr{O}_{\mathbb{P}(V)}(1)$, which comes from the fact that the total space of $\mathscr{O}_{\mathbb{P}(V)}(1)$ is the blowup of V at the origin, and the action on V lifts to an action on the blowup (since we blew up a fixed point).

Definition 2.52. Let X be finite type over an algebraically closed field k, G act on X, and $\mathscr{L}$ be a G-linearized line bundle on X. A point $x \in X(k)$ is *semistable* if there exists $s \in H^0(X, \mathscr{L}^n)$ for some positive integer n such that

(1) s is G-invariant under the natural action of G on $H^0(X, \mathscr{L}^n)$ induced by the linearization;
(2) $X_s = X \smallsetminus \mathbb{V}(s)$ is affine;
(3) $x \in X_s$.

Furthermore, x is *stable* if the action on X_s is closed.

Remarks 2.53.

(1) We see that the *semistable locus* $X^{\mathrm{ss}} \subset X$ is open.
(2) If $\mathscr{L}$ is ample, then X_s is automatically affine.

Theorem 2.54 (Geometric invariant theory)**.** *Let G be reductive and $(X, \mathscr{L})$ be a finite-type scheme over k with a G-linearized line bundle. Then the semistable locus*

$$X^{\mathrm{ss}}(\mathscr{L}) := \left\{ x \text{ is semistable with respect to } \mathscr{L} \right\} \subseteq X$$

is open and there exists a good quotient

$$\pi\colon X^{\mathrm{ss}}(\mathscr{L}) \to X^{\mathrm{ss}}(\mathscr{L}) /\!/ G = Y$$

and an open $\tilde{Y} \subset Y$ such that $\pi^{-1}(\tilde{Y}) = X^s(\mathscr{L})$ and $X^s(\mathscr{L}) \to \tilde{Y}$ is geometric. Moreover, there exists an ample line bundle $\mathscr{M}$ on Y and positive integer N such that

$$\pi^* \mathscr{M} = \mathscr{L}^{\otimes N}_{|X^{\mathrm{ss}}}.$$

Proof sketch. Write

$$X^{\mathrm{ss}}(\mathscr{L}) = X_{s_1} \cup \cdots \cup X_{s_n},$$

which requires replacing $\mathscr{L}$ by $\mathscr{L}^{\otimes N}$. Here, we have

$$X_{s_i} = \{x \in X \mid s_i(x) \neq 0\}$$

for $s_i \in H^0(X, \mathscr{L})^H$. By definition, each $X_{s_i} = \operatorname{Spec}\Gamma(X_{s_i}, \mathscr{O})$ is affine. Applying affine GIT, we obtain good quotients

$$\pi_i \colon X_{s_i} \to U_i = \operatorname{Spec}\Gamma(X_{s_i}, \mathscr{O})^G.$$

Our goal is now to glue the U_i. If we consider the function $\frac{s_i}{s_j}$, it is a G-invariant function on X_{s_j}, so it descends to a function σ_{ij} on U_j. Now $(U_j)_{\sigma_{ij}}$ is the categorical quotient of $X_{s_i} \cap X_{s_j}$, but so is $(U_i)_{\sigma_{ji}}$, so they are canonically isomorphic via

$$\psi_{ij} \colon (U_j)_{\sigma_{ij}} \simeq (U_i)_{\sigma_{ji}}.$$

These satisfy the cocycle condition because everything is a canonical isomorphism between categorical quotients, so the U_i glue to a scheme. The $\{\sigma_{ij}\}$ give a 1-cocycle defining a line bundle $\mathscr{M}$ such that $\pi^*\mathscr{M} = \mathscr{L}$. Ampleness follows from the fact that the distinguished opens U_i are affine. □

Theorem 2.55. *If X is projective and $\mathscr{L}$ is ample, then*

$$\pi \colon X^{\mathrm{ss}}(\mathscr{L}) \to Y = \operatorname{Proj} \bigoplus_{m \geq 0} H^0(X, \mathscr{L}^{\otimes m})^G$$

is a projective scheme and $\pi^*\mathscr{O}_Y(1) = \mathscr{L}^{\otimes N}$.

Remark 2.56. In this case, we have

$$X^{\mathrm{ss}}(\mathscr{L}) = \left\{x \in X \mid \text{there exists } \sigma \in H^0(X, \mathscr{L}^{\otimes N})^G, \sigma(x) \neq 0\right\}.$$

3. Stacks

3.1. **Quotient stacks.** Let X be a G-scheme for an affine group scheme G.

Definition 3.1. Define the *quotient stack*

$$[X/G] \colon \mathrm{Sch}_{/k} \to \mathrm{Gpd}$$

to be the contravariant (pseudo)functor defined by

$$[X/G](U) = \left\{\begin{matrix}\text{principal } G\text{-bundle } E \to U \\ \text{with a } G\text{-morphism } E \to X\end{matrix}\right\}.$$

Now we will work over a field k and require G to be reductive. We will need to make sense of everything in the above definition.

Definition 3.2. A morphism $\pi \colon E \to B$ is a *fibration* with fiber F if there exists a surjective étale (equivalently, smooth) morphism $B' \to B$ such that the diagram

$$\begin{tikzcd} F \times B' \arrow[r, "\simeq, \sigma"] \arrow[dr] & E \times_B B' \arrow[r] \arrow[d] & E \arrow[d] \\ & B' \arrow[r] & B \end{tikzcd}$$

is a Cartesian diagram.

Definition 3.3. Suppose that F and E are G-schemes. We say that a fibration $E \to B$ with fiber F is *G-equivariant* if σ in the previous definition is G-equivariant.

Definition 3.4. A morphism $\pi \colon E \to B$ is a *principal G-bundle* if it is a G-equivariant fibration with fiber G, where G acts on itself by left multiplication.

Lemma 3.5 (G reductive)**.** *Suppose that $\pi\colon E \to B$ is a principal G-bundle. Then*

(1) *$B \simeq E/G$ is a good geometric quotient;*
(2) *G acts freely on E in the stronger sense that*

$$\Psi\colon G \times E \to E \times E$$

is a closed embedding.

Proposition 3.6 (Converse)**.** *If G acts freely on X* ***and*** *the quotient X/G exists as a scheme, then $X \to X/G$ is a principal G-bundle.*

Since $\pi\colon E \to B$ is affine, we can glue affine GITs to get a good quotient $E/G = B^{\mathrm{GIT}}$. This produces the diagram

$$\begin{tikzcd}
G \times B' \arrow[r, "\simeq,\sigma"] \arrow[dr] & B' \times_B E \arrow[r] \arrow[d] & E \arrow[d] \arrow[dr] & \\
& B' \arrow[r] & B & B^{\mathrm{GIT}} \arrow[l] \arrow[dl] \\
& & (B')^{\mathrm{GIT}}, \arrow[ul, "\sim"] &
\end{tikzcd}$$

where the bottom square is Cartesian. This implies that $B^{\mathrm{GIT}} \to B$ is an isomorphism because this is local on the target.

An alternative proof is to consider the diagram

$$\begin{tikzcd}
G \times E \arrow[r, "\Psi"] & E \times_B E \arrow[r, hook] & E \times E \\
G \times E' \arrow[r, "\Psi'"] \arrow[u] & E' \times_{B'} E' \arrow[r, hook] \arrow[u] & E' \times E'. \arrow[u]
\end{tikzcd}$$

Because G acts freely on E', Ψ' is a closed embedding, and therefore so is Ψ. Then we can deduce things going in a direction that is not the one we usually go in.

Proposition 3.7. *Suppose G acts freely on X and a good geometric quotient $\pi\colon X \to X/G$ exists. Then π is a principal G-bundle.*

Exercise 3.8. Suppose Y is an affine G-scheme and $X \to B$ is a principal G-bundle. Then $Y \times_{\operatorname{Spec} k} X/G \to X/G \cong B$ is a G-equvariant fibration with fiber Y (here, G acts on the first factor Y).

Exercise 3.9. Let $f\colon X \to Y$ be a G-morphism of principal G-bundles. Then we have a Cartesian diagram

$$\begin{tikzcd}
X \arrow[r, "f"] \arrow[d] & Y \arrow[d] \\
X/G \arrow[r, "f/G"] & Y/G.
\end{tikzcd}$$

Hint: consider the diagram

$$\begin{tikzcd} X \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\ & \square \arrow[r] \arrow[d] & Y \arrow[d, "\pi_Y"] \\ & B' \arrow[r] & B \end{tikzcd}$$

and use the next exercise to conclude that the dashed arrow is an isomorphism.

Exercise 3.10. A G-equivariant morphism of principal G-bundles over some base is an isomorphism.

Example 3.11. Let X be a G-scheme with G affine. Define the *quotient stack*

$$[X/G]\colon \mathrm{Sch}_{/k} \to \mathrm{Gpd}$$

to be the contravariant (pseudo)functor given by

$$[X/G](U) = \left\{ \begin{matrix} \text{principal } G\text{-bundle } E \to U \\ \text{with a } G\text{-morphism } E \to X \end{matrix} \right\}.$$

Note that this is a groupoid by the last exercise. The morphisms in this category are $f^*\colon [X/G](U) \to [X/G](U')$ for morphisms $f\colon U' \to U$, and are given by

$$\begin{tikzcd} E \arrow[r] \arrow[d] & X \\ U \end{tikzcd} \mapsto \begin{tikzcd} E \times_U U' \arrow[r] \arrow[d] & X \\ U'. \end{tikzcd}$$

We also have a natural isomorphism

$$\varepsilon_{f,g}\colon g^* \circ f^* \cong (f \circ g)^*$$

but do not have equality on the nose. This boils down to the fiber product being unique up to unique isomorphism.

Exercise 3.12. If X is a principal G-bundle, then $[X/G]$ is representable by X/G. *Hint: consider the Cartesian diagram*

$$\begin{tikzcd} E \arrow[r] \arrow[d] & X \arrow[d] \\ U \arrow[r] & B \end{tikzcd}$$

and show that $[X/G](U) = \mathrm{Mor}(U, B)$.

Let $H < G$ be a closed subgroup and Y be an H-scheme with a good quotient. Then H acts on $G \times Y$ by $h \cdot (g, y) = (gh^{-1}, hy)$, and this action **commutes** with the G-action on $G \times Y$. Therefore, G acts on

$$G \times_H Y := (G \times Y) /\!/ H = (G \times Y)/H$$

by the formula

$$g_1\overline{(g_2, y)} = \overline{(g_1 g_2, y)}.$$

In addition, the quotients become

$$(G \times_H Y) /\!/ G = Y /\!/ H.$$

To see this, suppose $X = G \times Y = \operatorname{Spec} A$ is affine. Then we have

$$\begin{aligned} X /\!/ (G \times H) &= \operatorname{Spec} A^{G \times H} \\ &= \operatorname{Spec}(A^G)^H \\ &= (X /\!/ G) /\!/ H \\ &= Y /\!/ H. \end{aligned}$$

In particular, we see that every H-quotient is a G-quotient.

Remark 3.13. If $Y \hookrightarrow X$ where X is a G-scheme, we have a natural map

$$G \times_H Y \to X \qquad \overline{(g, y)} \mapsto gy$$

whose image is GY.

Let X and Y be G-schemes admitting good quotients.

Definition 3.14. A G-morphism $f \colon X \to Y$ is *strongly étale* if in the diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\pi_X"] & Y \arrow[d] \\ X /\!/ G \arrow[r, "f/G"] & Y /\!/ G \end{tikzcd}$$

the morphisms π_X and f induce an isomorphism $X \simeq (X /\!/ G) \times_{Y /\!/ G} Y$ **and** f/G is étale.

Example 3.15. Consider $\mathbb{C}^\times \times \mathbb{C}^\times \subset \mathbb{C}^\times \times \mathbb{C} \subset \mathbb{C} \times \mathbb{C}$. Consider the action of $G = \mathbb{C}^\times$ by $\lambda \cdot (x, y) = (\lambda x, \lambda y)$. The quotient of $\mathbb{C} \times \mathbb{C}$ is a point, the quotient of $\mathbb{C}^\times \times \mathbb{C}$ is $\mathbb{C} = \operatorname{Spec} \mathbb{C}\left[\frac{1}{y}\right]$, and the quotient of $\mathbb{C} \times \mathbb{C}^\times$ is $\mathbb{C}^\times$. In the diagram

$$\begin{tikzcd} \mathbb{C}^\times \times \mathbb{C}^\times \arrow[r, hook] \arrow[d] & \mathbb{C}^\times \times \mathbb{C} \arrow[r, hook] \arrow[d] & \mathbb{C} \times \mathbb{C} \arrow[d] \\ \mathbb{C}^\times \arrow[r, hook] & \mathbb{C} \arrow[r] & \mathrm{pt}, \end{tikzcd}$$

we see that the square on the left is Cartesian, but the square on the right is not. Thus $\mathbb{C}^\times \times \mathbb{C}^\times \hookrightarrow \mathbb{C}^\times$ is strongly étale, but $\mathbb{C}^\times \times \mathbb{C} \hookrightarrow \mathbb{C} \times \mathbb{C}$ is not.

Lemma 3.16. *If f is strongly étale, then*

(1) *f is étale;*
(2) *$f(X)$ is a saturated open in Y, or namely,*
$$f(X) = \pi_Y^{-1}(f/G(X /\!/ G)).$$
(3) *For all closed points $\bar{x} \in X /\!/ G$, we have $\pi_X^{-1}(\bar{x}) \simeq \pi_Y^{-1}(f/G(\bar{x}))$.*
(4) *The morphism $f_{|G \cdot x} \colon G \cdot c \to G \cdot f(x)$ is an isomorphism for all $x \in X(k)$ and $G \cdot x$ is closed in X if and only if $G \cdot f(x)$ is closed in Y. Also, we have $G_x \simeq G_{f(x)}$.*
(5) *X is normal at x.*

Lemma 3.17 (Luna's fundamental lemma)**.** *Suppose $f \colon X \to Y$ is a G-morphism of affine G-schemes. Suppose that $x \in X(k)$ is a closed point such that*

(1) *$T = G \cdot x$ is closed;*
(2) *The morphism $f_{|T} \colon T \to Y$ is injective (equivalently, $G_x \simeq G_{f(x)}$);*
(3) *f is étale at x.*

Then there exists a saturated affine open $U \subset X$ such that $V = f(U)$ is a saturated open affine in Y, $T \subset U$, and $f_{|U} \colon U \to V$ is strongly étale.

Theorem 3.18 (Luna's étale slice theorem)**.** *Suppose X is an affine G-scheme and $x \in X(k)$ is such that $G \cdot x$ is closed. Then there exists a G_x-invariant locally closed affine $V \subset X$ with $x \in V$ such that $U := G \cdot V$ is a G-saturated open affine in X and the morphism*

$$G \times_{G_x} V \twoheadrightarrow U$$

is strongly étale. Moreover, if $x \in X$ is a smooth point, then we have a strongly étale morphism

$$\eta \colon V \to N_{(G \cdot x/X),x} = T_{X,x}/T_{G \cdot x,x}$$

which is G_x-equivariant such that $\eta(x) = 0$.

In other words, there is a diagram

$$\begin{tikzcd}
G \times_{G_x} N \arrow[d] & G \times_{G_x} V \arrow[l, "\text{strongly étale}"'] \arrow[r, "\text{strongly étale}"] \arrow[d] & U \arrow[r, hook] \arrow[d] & X \arrow[d] \\
N /\!/ G_x & G \times_{G_x} V /\!/ G = V /\!/ G_x \arrow[l, "\text{étale}"'] \arrow[r, "\text{étale}"] & U /\!/ G \arrow[r, hook] \arrow[ll, dashed, bend left] & X /\!/ G
\end{tikzcd}$$

where all of the squares are Cartesian.

Corollary 3.19. *Suppose X is an affine G-scheme. Let $X_0 \subset X$ be the subset of points such that*

(1) *$G \cdot x$ is closed;*
(2) *$G_x \simeq 1$.*

Then X_0 is a saturated open and $X_0 \to X_0 /\!/ G$ is a principal G-bundle.

Proof. Let $x \in X_0$ such that $G_x \simeq 1$ and $G \cdot x$ is closed. By Luna's étale slice theorem, there is a strongly étale morphism

$$\begin{tikzcd}
G \times V \arrow[r, equal] \arrow[dr, "\text{trivial}"'] & G \times_{G_x} V \arrow[r, "\text{strongly}", "\text{étale}"'] \arrow[d] & U \arrow[r, hook, "\text{saturated}"'] \arrow[d] & X \\
& V \arrow[r, "\text{étale}"] & U /\!/ G,
\end{tikzcd}$$

and covering X_0 by such U gives the desired result. □

Corollary 3.20. *Suppose X is a smooth affine G-scheme. Then X^G is smooth.*

Proof. Let $x \in X^G$. Then $G \cdot x = \{x\}$ is closed and $G_x = G$. By Luna's étale slice theorem, we have

$$\begin{tikzcd}
T_{X,x} \arrow[d] & G \times_G V = V \arrow[l, "\text{strongly}", "\text{étale}"'] \arrow[r, "\text{strongly}", "\text{étale}"'] \arrow[d] & U \arrow[d] \\
T_{X,x} /\!/ G & V /\!/ G \arrow[l] \arrow[r] & U /\!/ G.
\end{tikzcd}$$

This gives us a correspondence

$$T_{X,x}^G \leftarrow (G \times_G V)^G \to U^G$$

of étale morphisms, and $T^G_{X,x}$ is a vector space, and hence smooth, so everything else is smooth. □

Lemma 3.21. *Let X be affine and $x \in X$ be such that G_x is reductive. Then there exists an étale G_x-morphism $\phi\colon X \to T_{X,x}$ such that $\phi(x) = 0$.*

Proof. Let $X = \operatorname{Spec} A$ with x corresponding to a maximal ideal $\mathfrak{m}$. Then of course we have

$$T_{X,x} = \operatorname{Spec} \operatorname{Sym}^\bullet(\mathfrak{m}/\mathfrak{m}^2).$$

There is a section $\mathfrak{m}/\mathfrak{m}^2 \to \mathfrak{m} \subset A$ which then lifts to an algebra morphism. This is clearly an isomorphism on tangent spaces, so it is étale. □

Lemma 3.22. *Every affine finite-type G-scheme X can be embedded as a G-stable closed subscheme of an affine space with G-action.*

Proof. If $X = \operatorname{Spec} A$ and A is finitely generated, then we choose G-invariant generators $x_1, \dots, x_n$ of A. □

Proof sketch of Luna's étale slice theorem. Suppose that $x \in X$ has a closed orbit $G \cdot x$. Then there is a closed immersion

$$G/G_x \simeq G \cdot x \hookrightarrow X$$

and thus the orbit is affine. Matsushima's criterion states that if G is reductive and $H \subset G$ is a closed subgroup, then G/H is affine if and only if H is reductive. Therefore, G_x is reductive. By Lemma 3.21, there is an étale G_x-morphism $\phi\colon X \to T_{X,x}$ such that $\phi(x) = 0$. Assume now that $x \in X$ is a smooth point. Then we can decompose

$$T_{X,x} = T_{G\cdot x,x} \oplus N_x,$$

where N_x is the normal space to $G{\cdot}x$ at x. Let $Y = \phi^{-1}(N_x)$. This is a closed subscheme of X and is smooth at x. We now consider the morphism

$$\psi\colon G \times_{G_x} Y \to X \qquad z := \overline{(1_G, x)} \mapsto x.$$

Then in fact $G \times_{G_x} Y$ is smooth at z (which checks part 4 of Luna's fundamental lemma). Now, note that

$$G \times Y \to G \times_{G_x} Y \qquad (1_G, x) \mapsto z$$

is a principal G_x-bundle. We also compute $G_z = G_{\overline{(1_G,x)}} \simeq G_x$ because $g \cdot \overline{(1_G, x)} = \overline{(1_G, x)}$ if and only if $(g, gx) \in G_x \cdot (1_G, x)$, which is equivalent to $g \in G_x$. Therefore, ψ is étale at z. This implies that $G \cdot z$ is closed in $G \times_{G_x} Y$ because $G_x \cdot x$ is closed in Y.

We can now apply Luna's fundamental lemma to $\psi\colon G \times_{G_x} Y \to X$ to obtain a saturated open $U \subset G \times_{G_x} Y$ such that

$$\psi|_U\colon U \to \psi(U) \subset X$$

is strongly étale. Now consider the diagram

$$\begin{tikzcd} G \times_{G_x} Y /\!/ G \arrow[r, "\simeq"] & Y /\!/ G_x \\ G \times_{G_x} Y \arrow[u] \arrow[r] & Y. \arrow[u] \end{tikzcd}$$

Because U is a saturated open in $G \times_{G_x} Y$, there exists a saturated open $V \subset Y$ such that $U = G \times_{G_x} V$. Therefore, we obtain a strongly étale morphism $G \times_{G_x} V \to X$.

This finishes the proof in the case when $x \in X$ is a smooth point. The general case follows by Lemma 3.22. $\square$

One application of Luna's étale slice theorem is that the quotient stack $\mathfrak{X} = [X/G]$ is locally isomorphic to $[V/G_x]$ near a closed point $x \in \mathfrak{X}$ (at least at a smooth point).

3.2. **General stacks.** Every stack is going to be a category fibered in groupoids. We begin by discussing some examples.

Example 3.23. Every contravariant functor (presheaf) $F\colon \mathrm{Sch}_S^{\mathrm{op}} \to \mathrm{Set}$ defines a category fibered in groupoids $\mathscr{G}_F$ over Sch_S whose objects are simply

$$\mathrm{Obj}(\mathscr{G}_F) = \Big\{(B, \beta) \mid \beta \in F(B)\Big\}$$

and whose morphisms are given by

$$\mathrm{Mor}(\mathscr{G}_F) = \Big\{(B', \beta') \to (B, \beta) \mid B' \xrightarrow{f} B, \beta' = F(f)(\beta)\Big\}.$$

The reason it is called a category fibered in groupoids is that there is a morphism to Sch_S and each $p^{-1}(B) = F(B)$ is a set (which is the simplest example of a groupoid).

Example 3.24. Consider the moduli problem $\mathscr{M}_g$, which is given by

$$\mathscr{M}_g(B) = \Big\{\text{isomorphism classes of flat families of genus } g \text{ smooth curves over } B\Big\}.$$

Morphisms are given by Cartesian diagrams

$$\begin{tikzcd} \mathscr{C}' \arrow[r] \arrow[d] & \mathscr{C} \arrow[d] \\ B' \arrow[r] & B, \end{tikzcd}$$

and over the identity arrow of B we clearly only have isomorphisms.

Example 3.25. If Z is a scheme, its functor of points $F = h_Z$ is a category fibered in groupoids.

If we have a category fibered in groupoids $p\colon \mathscr{C} \to \mathrm{Sch}_S$, we have the Isom functor

$$\mathrm{Isom}(X, Y)\colon \mathrm{Sch}_{/B}^{\mathrm{op}} \to \mathrm{Set} \qquad \mathrm{Isom}(X, Y)(B') = \left\{\begin{matrix}\text{isomorphisms between} \\ f^*X \text{ and } f^*Y\end{matrix}\right\}$$

for any two objects $X, Y \in \mathscr{C}(B)$.

Example 3.26. If $\mathscr{C} = \mathscr{C}_F$ for a functor $F\colon \mathrm{Sch}_S^{\mathrm{op}} \to \mathrm{Set}$, then

$$\mathrm{Isom}_B(X, Y)(B') = \begin{cases} \{1\} & f^*X = f^*Y \\ \varnothing & f^*X \neq f^*Y. \end{cases}$$

Definition 3.27. A *stack* is a category $\mathscr{C}$ fibered in groupoids over Sch_S if

(1) For all $B \in \mathrm{Sch}_S$, the Isom functor $\mathrm{Isom}(X, Y)$ is a sheaf on the big étale site of B;

(2) Descent is effective in the étale topology for all B. More precisely, given a surjective étale cover $B' = \bigsqcup B_i \to B$, $x_i \in \mathscr{C}(B_i)$, and isomorphisms $\phi_{ij} \colon x_i|_{B_i \times_B B_j} \simeq x_j|_{B_i \times_B B_j}$ satisfying the cocycle condition $\phi_{ij} \circ \phi_{jk} = \phi_{jk}$ on triple overlaps $B_i \times_B B_j \times_B B_k$, there exists $x \in \mathscr{C}(B)$ such that $x_i \simeq x|_{B_i}$.

Equivalently, it is a sheaf of groupoids on $(\mathrm{Sch}_S)_{\acute{e}t}$.

Definition 3.28. The *big {étale, smooth, fppf, Zariski, etc} site* is the category Sch_S with the Grothendieck topology in which covers are S-morphisms which are a disjoint union of {étale, smooth, fppf, open immersions, etc} morphisms and jointly surjective. This is denoted by $(\mathrm{Sch}_S)_{\text{ét,smooth,fppf,etc}}$.

Definition 3.29. A *sheaf* on $(\mathrm{Sch}_S)_\tau$ is a presheaf F satisfying the sheaf condition that

$$F(U) \to F(U') \rightrightarrows F(U' \times_U U')$$

is an equalizer for all τ-covers $U' \to U$.

Example 3.30. Because schemes are sheaves in the fpqc topology, every scheme is a stack.

Now suppose that $\mathfrak{X}$, $\mathfrak{Y}$, and $\mathfrak{Z}$ are stacks with morphisms $f \colon \mathfrak{X} \to \mathscr{Z}$ and $g \colon \mathfrak{Y} \to \mathscr{Z}$. We can form the fiber product

$$\mathfrak{X} \times_{\mathfrak{Z}} \mathfrak{Y}(B) = \Big\{(x, y) \mid x \in \mathfrak{X}(B), y \in \mathfrak{Y}(B), f(x) \simeq g(y)\Big\}.$$

The main point is that this fiber product is again a stack.

Definition 3.31. A morphism of stacks $f \colon \mathfrak{X} \to \mathfrak{Y}$ is *representable by schemes* if for every scheme B and morphism $B \to \mathfrak{Y}$, the fiber product $\mathfrak{X} \times_{\mathfrak{Y}} B$ is a scheme.

Example 3.32. The morphism $X \to [X/G]$ given by the action

$$\begin{tikzcd} G \times X \arrow[r, "\sigma"] \arrow[d] & X \\ X & \end{tikzcd}$$

is representable by schemes because for any

$$\begin{tikzcd} E \arrow[r] \arrow[d] & X \\ B & \end{tikzcd}$$

giving a morphism $B \to [X/G]$, the diagram

$$\begin{tikzcd} X \arrow[r] & {[X/G]} \\ E \arrow[r] \arrow[u] & B \arrow[u] \end{tikzcd}$$

is Cartesian.

Definition 3.33. An *algebraic space* is a sheaf F of sets on $(\mathrm{Sch}_X)_{\acute{e}t}$ such that

(1) The diagonal morphism $\Delta_F \colon F \to F \times F$ is representable by schemes (this implies that any morphism from a scheme is representable by schemes because $B \times_F C = (B \times C) \times_{F \times F} F$, and the two conditions are in fact equivalent);

(2) There exists a surjective étale morphism $U \to F$ where U is a scheme.

Definition 3.34. A stack $\mathfrak{X}$ is *Deligne-Mumford* (*resp. algebraic*) if

(1) The diagonal morphism $\Delta_{\mathfrak{X}} \colon \mathfrak{X} \to \mathfrak{X} \times \mathfrak{X}$ is representable by algebraic spaces;[2]
(2) There exists a surjective étale (resp. smooth) morphism $U \to \mathfrak{X}$ where U is a scheme.

Example 3.35. The quotient stack $[X/G]$ is an algebraic stack and the morphism $X \to [X/G]$ is a smooth atlas (at least in characteristic zero) and an algebraic space if G is finite and all stabilizers are trivial. Without the assumption on stabilizers, it is a Deligne-Mumford stack.

Because étale descent is effective for principal G-bundles, we will not discuss descent. The reason that $X \to [X/G]$ is representable is because the square

$$\begin{tikzcd} E \arrow[r] \arrow[d] & X \arrow[d] \\ B \arrow[r] & {[X/G]} \end{tikzcd}$$

is Cartesian, and we see that E is a scheme. Here, if we have morphisms $D \to B$ and $D \to X$, the fact that $X \times_{[X/G]} X \simeq G \times X$ implies that $E \times_B D \cong G \times D$ and gives us a morphism $D \to E$, which is the desired result. The morphism $X \to [X/G]$ is also clearly smooth and surjective, so it is a smooth atlas.

Example 3.36. If we define $\mathscr{M}_g$ to send a scheme B to the **set** of isomorphism classes of flat families $\mathscr{C} \to B$ of genus g smooth curves over B, then it is not a stack. For example, consider a genus g curve C with a nontrivial involution and an étale double cover $B' \to B$. Then we can consider the quotient X of $C \times B'$ by the diagonal action of $\mathbb{Z}_2$, and the resulting diagram

$$\begin{tikzcd} C \times B' \arrow[r] \arrow[d] & X \arrow[d] \\ B' \arrow[r] & B \end{tikzcd}$$

is Cartesian, but $X \neq C \times B$ in general. This shows that the Isom functor $\mathrm{Isom}(C \times B, X)$ is not a sheaf and that descent is not effective.

However, if we allow the total space to be an algebraic space, then if we consider the diagram

$$\begin{tikzcd} & \mathscr{C}' \arrow[d] & \mathrm{pr}_1^* \mathscr{C}' \arrow[l] \arrow[r, "\sim"] \arrow[d] & \mathrm{pr}_2{}^* \mathscr{C}' \arrow[dl] \\ B & B' \arrow[l] & B' \times_B B', \arrow[l, shift right, "\mathrm{pr}_1"'] \arrow[l, shift left, "\mathrm{pr}_2"] & \end{tikzcd}$$

then taking the pushout in the category of algebraic spaces gives us a family $\mathscr{C}$ over B filling in the diagram.

Example 3.37. Let X be the affine line with doubled origin and $\mathbb{Z}_2$ act by switching the two origins and exchanging $x \leftrightarrow -x$ for $x \neq 0$. Then $[X/\mathbb{Z}_2]$ is a sheaf on $\mathrm{Sch}_{\text{ét}}$ (and in fact an algebraic space) but is not a scheme because the two origins do not lie in a single affine.

[2]Some references require the diagonal to be quasicompact and separated, but this is not necessary.

Example 3.38. Consider two copies of $\mathbb{P}^1$ which are glued along two nodes. This configuration admits two inclusions f, g into $\mathbb{P}^2$ (where one becomes a line and the other a conic), and there is a pushout $\mathbb{P}^2 \cup_{f=g} \mathbb{P}^2$, which is not projective. There is an involution induced by switching the two copies of $\mathbb{P}^1$ and the two nodes, and the stack $[X/\mathbb{Z}_2]$ is not a scheme because the two nodes do not lie in a single affine, but is an algebraic space.

Remark 3.39. If we have a presentation $U \to \mathfrak{X}$, then we can replace the condition that the diagonal is representable by requiring that the presentation is representable.

3.3. **Equivalence relations.** It is easier to deal with morphisms of schemes than morphisms of stacks, so we will give another way to think about stacks.

Definition 3.40. A *pre-relation* on a scheme U over S is a morphism $j\colon R \to U \times_S U$ where R is a scheme. It is a *relation* if j is a monomorphism (equivalently, $\Delta\colon R \to R \times_{U\times U} R$ is an isomorphism, or the morphism is injective on T-points for all T).

A *pre-equivalence* (*pre-*)*relation* is a (pre-)relation such that the image of $R(T)$ in $U(T) \times U(T)$ is an equivalence relation for all T. An *equivalence relation* is a pre-equivalence **relation**.

Example 3.41. Let $G \times X \xrightarrow{\sigma} X$ be a group action and

$$\Psi\colon G \times X \xrightarrow{(\sigma, \mathrm{pr}_2)} X \times X.$$

Then Ψ is a pre-equivalence pre-relation and is an equivalence relation if all stabilizers are trivial.

Definition 3.42. A *groupoid scheme* over S is a quintuple (U, R, s, t, c) **and**

(1) An S-morphism $j\colon R \xrightarrow{(t,s)} U \times_S U$;
(2) An S-morphism $c\colon R \times_{t,U,s} R \to R$

such that $j(T)$ and $c(T)$ define a groupoid on $U(T)$.

Exercise 3.43. If (U, R, s, t, c) is a groupoid scheme, then $j\colon R \to U \times U$ is a pre-equivalence pre-relation.

Exercise 3.44. Every equivalence relation $j\colon R \to U \times U$ extends uniquely to a groupoid scheme (U, R, s, t, c). *Hint: define c using the Yoneda lemma.*

Example 3.45. Let $f\colon U \to \mathfrak{X}$ be a presentation of an algebraic space $\mathfrak{X}$. Then if we set $R = U \times_{\mathfrak{X}} U$, the morphism $j\colon R \to U \times U$ is an **étale** equivalence relation. Here, the set

$$R(T) = \left\{(\alpha, \beta) \mid \alpha, \beta \in U(T), f \circ \alpha = f \circ \beta \in \mathfrak{X}(T)\right\}$$

is obviously an equivalence relation.

Our goal now is to discuss quotients by pre-relations $j\colon R \to U \times U$. If T is a scheme, we get an induced equivalence relation $\sim_T$ on $U(T)$. In particular, we obtain a presheaf on $(\mathsf{Sch}_{/S})_{\acute{e}t}$ given by

$$T \mapsto U(T)/\sim_T .$$

Definition 3.46. The quotient U/R is the sheafification of this presheaf.

Remark 3.47. If we consider U and R as sheaves, then the diagram

$$\begin{tikzcd} R \arrow[r, shift left, "s"] \arrow[r, shift right, "t"'] & U \arrow[r] & U/R \end{tikzcd}$$

is a coequalizer in the category of sheaves.

Remark 3.48. For any surjection of sheaves on any site $q\colon \mathscr{F} \to \mathscr{G}$, the diagram

$$\begin{tikzcd} \mathscr{F} \times_{\mathscr{G}} \mathscr{F} \arrow[r] \arrow[d] & \mathscr{F} \arrow[d, "q"] \\ \mathscr{F} \arrow[r, "q"] & \mathscr{G} \end{tikzcd}$$

is both a pullback and a pushout.

If we apply this to the diagram

$$\begin{tikzcd} R \arrow[dr] \arrow[drr, bend left, "s"] \arrow[ddr, bend right, "t"'] & & \\ & U \times_{U/R} U \arrow[r] \arrow[d] & U \arrow[d] \\ & U \arrow[r] & U/R, \end{tikzcd}$$

then the map $R \to U \times_{U/R} U$ is surjective as a map of sheaves.

Theorem 3.49. *Let U be a scheme and $j\colon R \to U \times U$ be an étale* ***equivalence relation****. Then the sheaf U/R is an algebraic space and $q\colon U \to U/R$ is a representable étale surjective morphism.*

Proposition 3.50. *Suppose $F\colon (\mathrm{Sch}_{/S})^{\mathrm{op}}_{\acute{e}t} \to \mathrm{Set}$ is an algebraic space with an étale atlas $f\colon U \to F$. Write $R = U \times_F U$. Then*

1. *The map $j\colon R \to U \times U$ is an equivalence relation;*
2. *The maps $s, t = \mathrm{pr}_1, \mathrm{pr}_2$ are étale and surjective;*
3. *The diagram $R \rightrightarrows U \to F$ is a coequalizer, and in particular $F \simeq U/R$.*

Proof. Note that the set

$$R(T) = \left\{(\alpha, \beta) \in U(T) \times U(T) \mid f \circ \alpha = f \circ \beta \in F(T)\right\}$$

defines an equivalence relation. Because objects in F have no automorphisms, we see that $R = U \times_F U \to U \times U$ is a monomorphism, so R is an equivalence relation.

The fact that s and t are étale is just the fact that the property of being étale is stable under base change.

Finally, the property of being a coequalizer is a property of any surjection of sheaves. □

Definition 3.51. Let $\mathfrak{X}$ be an algebraic (resp. DM) stack. A morphism $\phi\colon \mathfrak{X} \to X$ to an algebraic space X is a *good* (*resp. coarse*) *moduli space* if

1. $\phi_*\colon \mathrm{QCoh}(\mathfrak{X}) \to \mathrm{QCoh}(X)$ is exact (namely ϕ is *cohomologically affine*);
2. For every algebraically closed field k, ϕ induces a bijection from closed k-points in $\mathfrak{X}$ to $X(k)$.[3]

[3] If $\mathfrak{X}$ is DM, then all points are closed, so this condition is equivalent to ϕ inducing a bijection from $\mathfrak{X}(k)$ to $X(k)$.

Remark 3.52. If ϕ exists, then it is universal among maps to algebraic spaces and hence unique.

Example 3.53. If G is reductive, then the map

$$\phi\colon \mathfrak{X} = [\operatorname{Spec} A/G] \to \operatorname{Spec} A^G = X$$

is a good moduli space in characteristic zero. The first follows from the fact that taking invariants is exact, and the second follows from GIT.

Theorem 3.54. *Suppose $\mathfrak{X} = [U/G]$ is a quotient stack where U is normal and G is reductive. If all stabilizers are finite, then $\mathfrak{X}$ is a Deligne-Mumford stack and has a coarse moduli space.*

For example, $\bar{\mathscr{M}}_g$ is of this form (where U is a suitable locus inside a Hilbert scheme of $\mathbb{P}^N$ and $G = \mathrm{PGL}_{N+1}$), so this theorem applies to it.

Proof. By Luna's étale slice theorem, there exists a strongly étale morphism $V \simeq V_0 \times_{G_x} G \xrightarrow{\psi} U$ where V_0 is affine and locally closed in U, $\psi(V)$ is saturated, and $G \cdot x \subset V$.

This implies that the morphism $[V_0/G_x] \simeq [V/G] \to [U/G]$ is strongly étale. Note that $[V_0/G_x]$ has a coarse moduli space $\operatorname{Spec} H^0(V_0, \mathscr{O}_{V_0})^{G_x}$. Therefore, there exists a strongly étale morphism $\mathfrak{Y} \to \mathfrak{X}$ where $\mathfrak{Y}$ has a coarse moduli space Y. Now consider the fiber product $\mathfrak{Y} \times_{\mathfrak{X}} \mathfrak{Y}$, which has a coarse moduli space Y_2. Then there are two morphisms $s, t\colon Y_2 \to Y$ which are affine and étale, so the left square in the diagram

$$\begin{tikzcd}
\mathfrak{Y} \times_{\mathfrak{X}} \mathfrak{Y} \arrow[r, shift left, "\mathrm{pr}_1"] \arrow[r, shift right, "\mathrm{pr}_2"'] \arrow[d] & \mathfrak{Y} \arrow[r, "\psi"] \arrow[d, "\phi"] & \mathfrak{X} \arrow[d] \\
Y_2 \arrow[r, shift left, "s"] \arrow[r, shift right, "t"'] & Y \arrow[r, dashed] & X
\end{tikzcd}$$

is Cartesian. Finally, $Y_2 \rightrightarrows Y$ is an étale equivalence relation, so we can form its quotient X, which is an algebraic space. $\square$

Remark 3.55. The theorem also applies in the case where G is reductive and all stabilizers are reductive.

Theorem 3.56 (Grothendieck)**.** *Let $f\colon T' \to T$ be an fppf morphism and $X' \to T'$ be separated and locally quasi-finite, then descent is effective.*